

From Spots to Cells: A Systematic Review of Computational Methods for Enhancing Spatial Transcriptomics Resolution to the Single-Cell Level

Yangji Chen,¹ Chunping Ouyang^{1,*} and Taolan Zhang^{2,*}

¹School of Computer Science, University of South China, Hengyang, 421001, China

²Health Management Center, The First Affiliated Hospital, Hengyang Medical School, University of South China, Hengyang, 421001, China

*Corresponding authors. ouyangcp@126.com (C. Ouyang); TaolZhan@usc.edu.cn (T. Zhang)

Abstract

Sequencing-based spatial transcriptomics platforms such as Visium capture transcriptome-wide expression but at spot-level resolution, where each measurement aggregates signals from multiple cells. Recovering single-cell information from these mixed observations is necessary for downstream analyses including cell–cell communication inference, niche characterisation, and biomarker discovery. Here we provide the first systematic review covering the complete computational pipeline from spots to cells, organised into four progressive stages: (I) spot-level deconvolution to estimate cell-type proportions, (II) single-cell mapping to assign reference cells to spatial coordinates, (III) pseudo single-cell reconstruction to synthesise per-cell expression profiles at sub-spot resolution, and (IV) multi-modal and image-guided enhancement using histology or paired modalities. We survey more than 25 representative methods published between 2020 and 2026, propose a two-axis taxonomy based on output resolution and information source, and provide a practical decision tree for method selection. We delineate what pseudo single-cell reconstruction can and cannot reliably infer, identify five open challenges, and discuss future directions including foundation models for spatial omics and the implications of emerging sub-cellular platforms.

Keywords spatial transcriptomics, cell-type deconvolution, single-cell mapping, pseudo single-cell reconstruction, Visium, multi-modal integration

Introduction

Spatial transcriptomics preserves the tissue coordinates of gene-expression measurements, allowing direct study of microenvironments, intercellular signalling, and spatially patterned disease processes that dissociated single-cell RNA sequencing (scRNA-seq) cannot access. Since its recognition as *Nature Methods*' Method of the Year in 2020 (Marx, 2021), the technology has become routine in both basic and translational research.

However, the most widely adopted spatial transcriptomics platform—10x Genomics Visium—operates at a resolution of 55 µm per spot, with each spot typically containing 1–10 cells. This “spot mixing” problem means that the measured expression at each location is a superposition of signals from multiple cells of potentially different types, obscuring the cell-type-specific biology that researchers seek to characterise. While newer platforms (Visium HD, Slide-seqV2, Stereo-seq) push toward higher resolution, and imaging-based methods (MERFISH, Xenium) achieve true single-cell or subcellular resolution, the vast majority of published spatial transcriptomics datasets—and the most cost-effective experimental designs—remain at spot-level resolution with transcriptome-wide gene coverage.

This gap has driven the development of computational methods that recover single-cell information from spot-level measurements. These methods vary in ambition: some estimate cell-type proportions within each spot, others place individual reference cells at spatial coordinates, and the most aggressive synthesise complete per-cell expression profiles at sub-spot resolution. They differ in their assumptions, data requirements, and the confidence warranted by their outputs.

Existing reviews cover parts of this space. Gaspard-Boulinc et al. (2025) provide a thorough survey of cell-type deconvolution but do not extend to mapping or reconstruction. Earlier reviews by Longo et al. (2021) and Tian et al. (2023) predate pseudo single-cell reconstruction methods entirely. No existing review treats the complete computational pipeline from spots to cells as a unified progression.

Contributions of this review. We provide the first systematic treatment of the full four-stage pipeline for enhancing spatial transcriptomics resolution:

1. **Stage I:** Spot-level deconvolution—estimating cell-type proportions per spot.
2. **Stage II:** Single-cell mapping—assigning reference cells to spatial coordinates.

3. **Stage III:** Pseudo single-cell reconstruction—synthesising per-cell expression at sub-spot resolution.
4. **Stage IV:** Multi-modal and image-guided enhancement, using histology and paired modalities.

We survey more than 25 methods published between 2020 and 2026, propose a two-axis taxonomy based on output resolution and information source, and provide a practical decision tree for method selection. Critically, we delineate what pseudo single-cell reconstruction can and cannot reliably infer (Box 1), addressing the growing concern that reconstructed cells are being treated as observational data in downstream analyses without appropriate caveats.

The remainder of this review is organised as follows. Section 2 introduces spatial transcriptomics platforms and formalises the resolution gap. Sections 3–6 present the four stages in detail. Section 7 provides unified comparison and practical guidance. Section 8 discusses open challenges, and Section 9 outlines future directions.

Background: Spatial Transcriptomics Platforms and the Resolution Gap

Spatial transcriptomics (ST) technologies measure gene expression while preserving the spatial context of cells within tissue (Ståhl et al., 2016). Over the past decade, two complementary platform families have emerged, distinguished by their measurement principle and the resulting trade-off between transcriptomic breadth and spatial resolution.

Sequencing-based platforms

Sequencing-based ST captures mRNA molecules via spatially barcoded oligonucleotide arrays, followed by next-generation sequencing. The 10x Genomics Visium platform—the most widely adopted—uses a regular grid of 55 μm -diameter spots spaced 100 μm apart, each capturing the transcriptomes of approximately 1–10 cells depending on tissue density. Slide-seqV2 reduces bead diameter to ~ 10 μm , approaching single-cell resolution in sparse tissues but still aggregating 1–3 cells in dense regions (Stickels et al., 2021). Stereo-seq achieves sub-micrometre capture resolution through DNA nanoball arrays (Chen et al., 2022), though computational binning is typically applied to reach sufficient sequencing depth. Most recently, Visium HD offers 2 μm bins that can in principle isolate individual cells, but the shallow per-bin coverage often necessitates computational aggregation back to larger units.

The defining advantage of sequencing-based platforms is *unbiased transcriptome-wide measurement*: all polyadenylated transcripts are captured without prior gene selection. The cost is spatial resolution—each measurement integrates signal from multiple cells, creating the “spot mixing” problem that motivates the computational methods reviewed here.

Imaging-based platforms

Imaging-based ST (MERFISH (Chen et al., 2015), seqFISH+, Xenium (Janesick et al., 2023), CosMx) detects individual RNA molecules via sequential rounds of fluorescent hybridisation or *in situ* sequencing, achieving true subcellular resolution. Each detected molecule is assigned to a segmented cell, yielding per-cell expression profiles without computational deconvolution.

The trade-off is gene-panel size: current commercial platforms measure 300–1,000 pre-selected genes (Xenium, CosMx) or up to $\sim 10,000$ with extended protocols (MERFISH). Panel design requires prior knowledge of which genes are informative, limiting discovery of unexpected biology. Additionally, imaging-based platforms are more expensive per unit area and require specialised instrumentation.

The resolution gap and the mixing model

For sequencing-based platforms, the fundamental computational challenge is that each spot’s expression vector $\mathbf{x}_s \in \mathbb{R}^G$ (where G is the number of genes) represents a superposition of contributions from N_s cells belonging to potentially different types:

$$\mathbf{x}_s = \sum_{i=1}^{N_s} \mathbf{g}_i + \boldsymbol{\varepsilon}_s, \quad (1)$$

where $\mathbf{g}_i$ is the true expression of cell i and $\boldsymbol{\varepsilon}_s$ captures technical noise (sampling, amplification, sequencing). Without additional information, recovering individual $\mathbf{g}_i$ from the aggregate $\mathbf{x}_s$ is an ill-posed inverse problem.

Three sources of external information are commonly exploited to regularise this problem: (i) a *single-cell RNA-seq reference* providing cell-type expression signatures $\boldsymbol{\mu}_k$; (ii) *histology images* providing cell counts, positions, and morphological features; and (iii) *spatial structure* encoding the expectation that neighbouring spots share similar compositions. The four computational stages reviewed in this paper differ in which information sources they exploit and what resolution of output they produce.

Role of the scRNA-seq reference

The majority of methods in Stages I–III require a paired or atlas-derived scRNA-seq reference. This reference serves multiple roles: defining the set of possible cell types $\{1, \dots, K\}$, providing expression signatures $\boldsymbol{\mu}_k$ for deconvolution, and supplying individual cell profiles for mapping or reconstruction.

However, the reference introduces its own biases. Technology effects (10x Chromium vs. Smart-seq2 vs. spatial capture) create systematic differences in gene detection rates and expression distributions. Donor and batch effects mean that a reference built from one individual may not represent the cell states present in another. Incomplete references—missing rare cell types or transient states—create blind spots that no computational method can overcome. These limitations motivate the development of reference-free approaches (Section 3) and highlight the importance of reference quality assessment before applying any resolution-enhancement method.

Stage I: Spot-Level Deconvolution

The first computational task in resolving spatial transcriptomics beyond spot level is *spot-level deconvolution*: estimating the proportion, and in some formulations the absolute abundance, of each cell type within every captured spot. Formally, the observed expression vector $\mathbf{x}_s$ at spot s is modelled as a mixture:

$$\mathbf{x}_s = \sum_{k=1}^K \pi_{sk} \boldsymbol{\mu}_k + \boldsymbol{\varepsilon}_s, \quad (2)$$

where π_{sk} denotes the proportion (or count) of cell type k , $\boldsymbol{\mu}_k$ is the reference expression signature, and $\boldsymbol{\varepsilon}_s$ captures noise. This system

Table 1 Comparison of spatial transcriptomics platforms

Platform	Resolution	Gene coverage	Throughput	Cost
<i>Sequencing-based</i>				
Visium	55 μm	Transcriptome-wide	High	Moderate
Slide-seqV2	10 μm	Transcriptome-wide	Moderate	High
Stereo-seq	0.5 μm	Transcriptome-wide	High	High
Visium HD	2 μm	Transcriptome-wide	High	High
<i>Imaging-based</i>				
MERFISH	Subcellular	300–10,000	Moderate	Very high
Xenium	Subcellular	300–5,000	Moderate	High
CosMx	Subcellular	1,000–6,000	Moderate	High
seqFISH+	Subcellular	10,000+	Low	Very high

is under-determined: the number of genes far exceeds the number of cell types, yet biological and technical confounders make naïve regression unreliable. Over the past five years, numerous methods have addressed this problem; we organise them into four families below.

Probabilistic and Bayesian frameworks

Probabilistic approaches treat deconvolution as a generative modelling problem, placing priors on cell-type abundances and expression profiles to regularise the solution and provide uncertainty estimates.

cell2location (Kleshchevnikov et al., 2022) employs a hierarchical Bayesian model that decomposes spot counts into contributions from cell types whose signatures are learned from a paired single-cell reference. By modelling both the reference uncertainty and the spatial data jointly, it yields posterior distributions over absolute cell counts per spot, enabling downstream spatial statistics that account for estimation uncertainty. Its scalability to Visium-scale datasets (thousands of spots, tens of cell types) has made it one of the most widely adopted tools.

Stereoscope (Andersson et al., 2020) formulates deconvolution as maximum-likelihood estimation under a negative-binomial observation model, sharing gene-specific dispersion parameters across spots. While it does not provide full posteriors, its parametric assumptions align well with count data and it performs competitively on tissues with moderate heterogeneity.

DestVI (López et al., 2022) extends the variational-inference framework of scVI to the spatial setting, learning a continuous latent space that captures both discrete cell-type identity and continuous within-type variation. This allows DestVI to model cell-state gradients—for example, activation states of immune cells—that purely discrete deconvolution methods cannot represent.

Probabilistic methods share the ability to propagate uncertainty into downstream analyses. Their main limitation is computational cost: full Bayesian inference (MCMC or structured variational families) scales poorly beyond ~ 50 cell types or $\sim 10,000$ spots without approximations.

Regression-based frameworks

Regression methods cast deconvolution as a constrained linear problem, solving Equation 2 directly with non-negativity and sum-to-one constraints.

RCTD (Robust Cell Type Decomposition) (Cable et al., 2022) fits a platform-effect-corrected Poisson regression per spot, using

a pre-computed reference built from annotated single-cell data. Its key innovation is explicit modelling of the platform effect between scRNA-seq and spatial technologies, which reduces systematic bias in proportion estimates. RCTD offers both “full” (all cell types) and “doublet” (at most two types per spot) modes, the latter being appropriate for high-resolution platforms such as Slide-seqV2 where spots are small.

CARD (Ma and Zhou, 2022) introduces spatial correlation into the regression framework by placing a conditional autoregressive prior on proportions across neighbouring spots. This encourages spatial smoothness—biologically motivated because cell-type distributions change gradually across tissue—and improves estimation in low-count spots by borrowing information from neighbours.

SpatialDWLS (Dong and Yuan, 2021) adapts the dampened weighted least-squares approach originally developed for bulk deconvolution, weighting genes by their discriminative power for each cell type. It is fast and deterministic but does not model spatial structure or provide uncertainty.

SPOTlight (Elosúa-Bayes et al., 2021) uses non-negative matrix factorisation (NMF) to first learn topic-like basis vectors from the reference, then projects spatial spots onto these bases. Its two-step design is computationally efficient but may lose information in the compression step.

Regression methods are generally faster than Bayesian alternatives and scale well, but they produce point estimates without uncertainty quantification and are sensitive to reference quality.

An important practical consideration for all regression methods is reference construction. The reference signature matrix should be built from well-annotated, technology-matched single-cell data with sufficient representation of all expected cell types. Under-representation of rare types (e.g., fewer than 50 cells) leads to unstable signature estimates and biased proportions. Several studies recommend filtering the reference to retain only marker genes with high specificity (fold-change > 1.5 , detection rate > 0.3 in the target type) to reduce noise from non-informative genes.

Reference-free approaches

When a matched single-cell reference is unavailable—common for rare tissues, archival samples, or non-model organisms—reference-free methods attempt to decompose spots using only the spatial data itself.

STdeconvolve (Hao et al., 2023) applies latent Dirichlet allocation (LDA), treating spots as “documents” and genes as “words”, to discover latent cell-type topics without external supervision. The number of topics K must be specified or selected via model-comparison criteria, and the resulting topics require post-hoc biological interpretation.

BayesTME extends the topic-model idea with a Bayesian spatial prior and explicit modelling of the tissue microenvironment, jointly inferring cell types and their spatial organisation. It can recover biologically meaningful components in well-structured tissues but struggles when cell types share highly overlapping transcriptomes.

Reference-free methods avoid the need for external data but face an inherent identifiability problem: without external anchoring, the decomposed components may not correspond to biologically defined cell types, and rare populations are easily absorbed into dominant topics.

Graph and deep-learning hybrids

Recent methods exploit graph neural networks or deep generative models to capture complex non-linear relationships between spots and cell types.

STdGCN constructs a spatial graph over spots and uses graph convolutional layers to propagate information across neighbours while predicting cell-type proportions. The graph structure naturally encodes tissue topology and allows the model to learn spatially varying deconvolution patterns.

DSTG (Deep Spatial and Temporal Graph) (Song and Su, 2021) similarly uses graph convolutions but incorporates semi-supervised learning from pseudo-spots generated by mixing reference cells in known proportions, providing training signal in the absence of ground-truth spatial labels.

GraphST (Long et al., 2023) combines graph attention networks with contrastive learning to jointly cluster spots and estimate compositions, exploiting the observation that spatially co-located spots tend to share similar cell-type mixtures.

These methods achieve strong performance on complex tissues but require GPU resources, careful hyperparameter tuning, and sufficient training data (real or simulated). Their “black-box” nature also limits interpretability compared to explicit probabilistic models.

A practical concern with graph-based methods is their sensitivity to graph construction parameters (number of neighbours k , edge-weighting scheme). Different choices of k can yield substantially different proportion estimates, particularly at tissue boundaries where the local neighbourhood assumption breaks down. Users should test multiple graph configurations and report sensitivity.

Benchmark evidence and selection guidance

Systematic benchmarks (Li et al., 2023; Yan et al., 2024) consistently show that no single method dominates across all tissue types and evaluation metrics. Key findings include: (i) probabilistic methods (cell2location, DestVI) tend to excel on complex tissues with many cell types; (ii) regression methods (RCTD, CARD) offer the best speed–accuracy trade-off for routine analyses; (iii) reference-free methods are viable only when reference data are truly unavailable and the tissue has well-separated cell types; and (iv) deep-learning methods show promise but lack the extensive validation of established tools.

For practical guidance, we recommend: use **cell2location** or **RCTD** as a default starting point when a good reference exists; add **CARD** when spatial smoothness is expected and the tissue is well-structured; consider **STdeconvolve** only as a complement or when no reference is available; and reserve graph-based methods for large datasets where their computational overhead is justified by improved performance on heterogeneous tissues.

Stage II: Single-Cell Mapping

Single-cell mapping addresses a complementary question to deconvolution: rather than estimating what fraction of each cell type occupies a spot, it asks *where* individual cells from a paired scRNA-seq reference should be placed in tissue space. The output is a set of spatial coordinates assigned to real (observed) single cells, producing a spatially resolved cell atlas without synthesising new expression profiles. This stage bridges deconvolution and full reconstruction by

providing a cell-level spatial layout that downstream analyses can use directly.

Tangram

Tangram (Biancalani et al., 2021) formulates mapping as a continuous optimisation problem. It learns a soft assignment matrix $\mathbf{M} \in \mathbb{R}^{C \times S}$ (cells $\times$ spots) that minimises the discrepancy between the predicted spatial expression—obtained by projecting single-cell profiles through $\mathbf{M}$ —and the observed spot expression. The objective is:

$$\min_{\mathbf{M} \geq 0} \|\mathbf{G}^\top \mathbf{M} - \mathbf{X}\|_F^2 + \lambda \mathcal{R}(\mathbf{M}), \quad (3)$$

where $\mathbf{G}$ is the single-cell expression matrix, $\mathbf{X}$ is the spatial expression matrix, and $\mathcal{R}$ is an optional density regulariser that encourages the number of cells mapped to each spot to match an external estimate (e.g., from nuclei segmentation). Tangram is flexible: it can operate in gene-space or in a reduced PCA space, and it accepts optional cell-density priors from histology. Its main limitation is that the soft assignment does not enforce a one-to-one mapping—a single reference cell can be fractionally assigned to multiple spots—which complicates biological interpretation.

CellTrek

CellTrek (Wei et al., 2022) takes a two-step approach. First, it co-embeds single cells and spatial spots into a shared latent space using mutual nearest neighbours, creating a joint UMAP representation. Second, it trains a random forest on the spatial coordinates of spots within this embedding and uses the forest to predict coordinates for each single cell. The predicted coordinates place cells at sub-spot resolution, producing a spatially resolved single-cell map.

CellTrek’s strength is its ability to leverage non-linear relationships captured by the random forest, potentially resolving fine spatial patterns that linear methods miss. However, the two-step design means errors in the co-embedding propagate into coordinate predictions, and the method does not enforce consistency with the observed spot-level expression (i.e., the sum of mapped cells’ expression need not equal the spot measurement).

CytoSPACE

CytoSPACE (Vahid et al., 2023) frames mapping as a linear assignment problem: given C reference cells and N spatial slots (where $N = \sum_s n_s$ and n_s is the estimated cell count in spot s), it finds the minimum-cost one-to-one assignment that minimises the total expression distance between each cell and its assigned spot. The assignment is solved efficiently using the shortest-augmenting-path algorithm.

The one-to-one constraint is CytoSPACE’s key advantage: every reference cell is placed exactly once, and every spatial slot receives exactly one cell, producing an unambiguous mapping. This makes downstream interpretation straightforward. The trade-off is rigidity: if the reference does not contain enough cells of a given type to fill all spots, the assignment is forced to use suboptimal matches. CytoSPACE therefore requires that the reference be sufficiently large and representative.

Limitations of mapping-based approaches

Despite their utility, mapping methods share several fundamental limitations:

1. **Reference dependence:** the spatial atlas can only contain cell types and states present in the reference. Novel populations or tissue-specific states absent from the scRNA-seq data will be missed entirely.
2. **Cell-count estimation:** all methods require an estimate of how many cells occupy each spot. Errors in this estimate—whether from image segmentation or from deconvolution—propagate directly into the mapping quality.
3. **Mapping ambiguity:** when multiple cell types have similar transcriptomes (e.g., fibroblast subtypes), the assignment becomes degenerate and the spatial placement is unreliable.
4. **No new expression information:** mapping places existing cells in space but does not refine or correct their expression profiles to account for spatial context. The mapped cells retain their original (non-spatial) expression values.
5. **Evaluation difficulty:** without ground-truth spatial single-cell data, it is hard to assess whether a mapping is biologically correct beyond checking aggregate consistency with spot-level measurements.

These limitations motivate Stage III methods, which go beyond placement to reconstruct spatially informed per-cell expression profiles.

Stage III: Pseudo Single-Cell Reconstruction

While deconvolution (Stage I) estimates proportions and mapping (Stage II) places reference cells in space, neither produces *de novo* per-cell expression profiles at sub-spot coordinates. Stage III methods—which we term *pseudo single-cell reconstruction*—aim to synthesise complete gene-expression vectors for individual cells within each spot, effectively transforming spot-level spatial transcriptomics into data that resemble single-cell resolution. This is the most ambitious and most easily over-claimed stage of the pipeline; we therefore devote particular attention to what these methods can and cannot reliably infer (Box 1).

The reconstruction task can be formalised as follows. Given a spot s with observed expression $\mathbf{x}_s$ and an estimated cell composition $\{(k_i, \hat{n}_i)\}_{i=1}^{N_s}$ from Stage I, the goal is to produce expression vectors $\{\hat{\mathbf{g}}_i\}_{i=1}^{N_s}$ such that $\sum_i \hat{\mathbf{g}}_i \approx \mathbf{x}_s$ and each $\hat{\mathbf{g}}_i$ is biologically plausible for cell type k_i . The problem is fundamentally under-determined: infinitely many decompositions satisfy the sum constraint. Methods differ in how they regularise the solution space.

Resampling and template-based reconstruction

The simplest strategy draws expression templates from the scRNA-seq reference and assigns them to spots proportionally.

Redeconve (Zhang et al., 2024) formulates reconstruction as sparse non-negative least squares (NNLS): for each spot, it selects a small subset of reference cells whose weighted combination best explains the observed spot expression. The sparsity constraint acts as a regulariser, favouring solutions that use few distinct profiles rather than diffuse mixtures. Redeconve is computationally efficient and deterministic, but its output is bounded by the diversity of the

reference—it cannot generate expression states not already present in the atlas.

SPROUT extends the resampling idea by incorporating spatial neighbourhood information: cells assigned to adjacent spots are encouraged to have correlated expression, reflecting the biological expectation that microenvironments vary smoothly. This spatial prior reduces artefactual discontinuities at spot boundaries but introduces a smoothing bias that may obscure sharp biological transitions (e.g., tumour–stroma interfaces).

Generative reconstruction

Generative methods learn a probabilistic model of single-cell expression and sample from it conditioned on the spatial observations.

SpatialScope (Shengquan et al., 2023) uses a diffusion-based generative model trained on the scRNA-seq reference. Given a spot’s observed expression and its deconvolution result, SpatialScope iteratively denoises latent representations to produce per-cell expression vectors that are both consistent with the spot total and lie on the learned single-cell manifold. The diffusion framework naturally handles multi-modality in expression space—for instance, generating distinct profiles for resting and activated T cells within the same spot—and provides a principled way to sample multiple plausible reconstructions, enabling uncertainty assessment.

Bulk2Space takes a related but simpler approach, training a variational autoencoder (VAE) on reference cells and then using the decoder to generate spatial single-cell profiles conditioned on deconvolution outputs. While faster than diffusion models, the VAE’s tendency toward posterior collapse can produce overly smooth reconstructions that underestimate biological variability.

Optimisation under reference and image constraints

The most recent methods combine multiple information sources—scRNA-seq reference, spatial expression, and histology images—in a joint optimisation framework.

CellMap (Li et al., 2026) formulates reconstruction as an optimal-transport problem: it seeks a mapping from reference cells to spatial positions that minimises a combined cost function incorporating expression similarity, spatial coherence, and (optionally) image-derived cell-density priors. The optimal-transport formulation guarantees a globally consistent assignment and naturally handles the constraint that the total reconstructed expression must match the observed spot signal.

CellRefiner (Wang et al., 2026) takes an iterative refinement approach. Starting from an initial reconstruction (e.g., from Redeconve or CytoSPACE), it uses H&E image segmentation to estimate per-spot cell counts and positions, then alternates between (i) updating expression assignments given positions and (ii) refining positions given expression, converging to a solution that is consistent with both molecular and morphological evidence. CellRefiner’s explicit use of histology makes it particularly effective on tissues where cell morphology is informative (e.g., distinguishing epithelial from stromal cells by nuclear shape).

Box 1: What pseudo single-cell reconstruction can and cannot reliably infer

Can reliably infer:

- Cell-type composition within each spot (given accurate Stage I input)
- Relative spatial distribution of cell types across tissue
- Co-occurrence patterns of cell types within microenvironments
- Ligand–receptor pair localisation to interacting cell types
- Spatial gene-expression gradients at sub-spot resolution
- Niche-level characterisation of tissue microenvironments

Cannot reliably infer:

- Absolute gene-expression values for individual cells
- Cell states absent from the scRNA-seq reference
- Stochastic gene-expression noise or burst kinetics
- Causal relationships between gene expression and spatial location
- True single-cell resolution measurements (these are model outputs, not observations)
- Rare cell-type contributions when <5% of spot composition

Best practice:

- Treat reconstructed cells as model outputs, not observational data
- Report reconstruction confidence scores
- Validate key findings against orthogonal data (IHC, smFISH)
- Propagate reconstruction uncertainty into downstream statistical tests
- Avoid causal claims based solely on reconstructed data

Validation against high-resolution ground truth

Evaluating pseudo single-cell reconstruction is inherently difficult because ground truth—true per-cell expression at known spatial coordinates—is rarely available. Current validation strategies include:

1. **Matched imaging-based ST:** comparing reconstructed profiles against MERFISH or Xenium measurements on serial sections from the same tissue. This provides the most direct validation but is limited to the gene panel of the imaging platform (typically 100–1,000 genes).
2. **In silico mixing benchmarks:** computationally aggregating annotated single cells into synthetic spots, running reconstruction, and measuring recovery of the original per-cell profiles. While controlled, these benchmarks do not capture real platform effects or tissue-specific confounders.
3. **Downstream biological consistency:** assessing whether reconstructed data recapitulate known biology—for example, whether ligand–receptor pairs are correctly localised to interacting cell types, or whether spatial gene-expression gradients match ISH (in situ hybridisation) patterns.

A major gap is the absence of standardised benchmark datasets and agreed-upon metrics for Stage III evaluation. Unlike deconvolution, where RMSE and JSD on proportions are widely accepted,

reconstruction quality involves both per-gene accuracy and global distributional fidelity, and no consensus metric has emerged.

We caution that reconstructed cells should be treated as *model outputs*, not observations. Downstream analyses—particularly differential expression testing and causal inference—must account for the additional uncertainty introduced by the reconstruction step. Reporting reconstruction confidence scores and validating key findings against orthogonal data (immunohistochemistry, smFISH) should be standard practice.

Application scenarios

Pseudo single-cell reconstruction has been applied to several biological questions where spot-level resolution is insufficient:

Tumour microenvironment dissection. In solid tumours, immune cells, fibroblasts, and malignant cells co-exist within individual Visium spots. Reconstruction enables spatial mapping of immune-excluded versus immune-infiltrated regions at cellular resolution, identification of immunosuppressive niches (e.g., regulatory T cells adjacent to tumour cells), and quantification of cell-type-specific ligand–receptor interactions that spot-level data conflate.

Developmental gradients. During organogenesis, progenitor cells differentiate along spatial axes. Reconstruction can resolve expression gradients within individual developmental zones (e.g., cortical layers, intestinal crypt–villus axis) that are blurred when multiple differentiation stages occupy the same spot. This enables spatial pseudotime analysis that combines trajectory inference with physical tissue coordinates.

Spatial gene regulatory networks. By providing per-cell expression estimates at known coordinates, reconstruction allows inference of spatially varying gene regulatory networks: identifying transcription factors whose target-gene activity differs between tissue compartments, even when the same cell type is present in both.

In all cases, conclusions drawn from reconstructed data should be validated against independent measurements. The biological plausibility of results does not guarantee their accuracy, and over-interpretation remains the primary risk of Stage III methods.

Stage IV: Multi-Modal and Image-Guided Enhancement

Spatial transcriptomics experiments routinely produce paired histology images (haematoxylin and eosin stains or immunofluorescence) that carry morphological information at sub-micrometre resolution. Stage IV methods use these complementary modalities, together with spatial coordinate structure, to refine or improve the outputs of earlier stages. Images provide cell-level information (counts, boundaries, morphology) that expression data alone cannot resolve; expression data provide molecular identity that images cannot.

Image segmentation as a cell-count prior

Accurate estimation of the number of cells per spot is a prerequisite for Stages II and III, yet it is rarely measured directly. Image segmentation fills this gap.

Cellpose (Stringer et al., 2021) is a generalist deep-learning segmentation model trained on diverse cell morphologies. Applied to H&E or DAPI-stained images co-registered with spatial transcriptomics slides, it produces per-spot nuclei counts that

serve as priors for deconvolution and mapping. Its gradient-flow representation generalises well across tissue types without retraining.

StarDist (Schmidt et al., 2018) uses star-convex polygons to represent nuclear boundaries, offering fast and accurate segmentation particularly suited to densely packed tissues (e.g., tumour cores) where touching nuclei are common. Its polygon representation naturally handles elongated or irregular nuclear shapes.

Mesmer (DeepCell) (Greenwald et al., 2022) is trained specifically on tissue microscopy and provides whole-cell (not just nuclear) segmentation when membrane markers are available, enabling more accurate cell-area estimation.

The segmentation-derived cell count $\hat{n}_s$ for each spot s directly informs the constraint $\sum_k \pi_{sk} = \hat{n}_s$ in deconvolution models and determines the number of slots in mapping algorithms like CytoSPACE.

Image-guided deconvolution

Beyond cell counting, histology images can inform the deconvolution process itself.

SegDecon integrates segmentation masks directly into the deconvolution objective: it partitions each spot into sub-regions corresponding to individual segmented cells, then estimates per-cell-type contributions within each sub-region rather than at the whole-spot level. This effectively increases the spatial resolution of deconvolution from spot-level ($\sim 55 \mu\text{m}$ for Visium) to cell-level ($\sim 10 \mu\text{m}$), bridging the gap between sequencing-based and imaging-based platforms.

BayesSpace (Zhao et al., 2022) takes a different approach: it applies a Bayesian spatial prior over a grid of sub-spot positions, using the spatial autocorrelation of expression to “enhance” resolution by a factor of $\sim 3\text{--}6\times$ without requiring image input. When histology is available, BayesSpace can incorporate image features as additional covariates, further improving sub-spot estimation.

Tangram with density input illustrates how existing Stage II methods benefit from image priors: by supplying Cellpose-derived cell densities as the regulariser $\mathcal{R}(\mathbf{M})$, Tangram’s mapping becomes more biologically grounded, particularly in regions where expression alone is ambiguous (e.g., stroma with mixed fibroblast subtypes).

Coordinate-aware smoothing

Even without explicit image input, the spatial coordinates of spots encode tissue structure that can regularise computational outputs.

Graph-based smoothing constructs a k -nearest-neighbour or Delaunay graph over spot positions and applies message-passing or diffusion operators to denoise expression estimates. Methods such as SpaGE and gimVI use this principle to impute missing genes in spatial data by transferring information from scRNA-seq references while respecting spatial locality.

Gaussian process (GP) regression offers a principled framework for spatial smoothing with uncertainty: by placing a GP prior over expression as a function of spatial coordinates, one obtains posterior mean estimates that are smooth where data are sparse and faithful to observations where data are dense. GP-based approaches are computationally expensive for large datasets but provide well-calibrated confidence intervals.

Integration with imaging-based ST

A growing experimental design pairs sequencing-based ST (e.g., Visium) with imaging-based ST (e.g., Xenium or MERFISH) on serial sections from the same tissue block. This creates an opportunity for computational integration:

- **Cross-platform anchoring:** using the shared gene panel (typically 100–500 genes measured by both platforms) to align coordinate systems and transfer cell-type labels from the imaging platform (which has true single-cell resolution) to the sequencing platform (which has transcriptome-wide coverage).
- **Gene imputation:** predicting the expression of genes measured only by sequencing-based ST at the single-cell positions identified by imaging-based ST, effectively combining the resolution of one with the breadth of the other.
- **Validation:** using imaging-based measurements as ground truth to evaluate and calibrate deconvolution or reconstruction outputs from the sequencing-based data.

This paired-platform strategy is increasingly adopted in atlas-scale projects (e.g., Human Cell Atlas) and offers a practical path toward single-cell-resolved, transcriptome-wide spatial data without requiring computational reconstruction to bear the full burden of resolution enhancement.

Unified Comparison and Practical Guidance

The preceding sections have presented methods organised by their position in the four-stage pipeline. Here we step back to provide a unified view that helps practitioners choose the right tool for their specific analytical context.

Two-axis taxonomy of methods

We propose a two-dimensional classification framework (Figure 3) that organises all reviewed methods along two orthogonal axes:

- **Output resolution** (vertical axis): what spatial granularity does the method produce? Options range from spot-level proportions (Stage I), through mapped cells at spot coordinates (Stage II), to per-cell expression at sub-spot positions (Stage III), and enhanced sub-spot estimates (Stage IV).
- **Information source** (horizontal axis): what external data does the method require? Categories include reference-only (scRNA-seq), image-only (histology), reference + image, coordinate-aware (spatial prior without image), and reference-free.

This taxonomy reveals several patterns. First, higher output resolution generally requires more information sources: Stage III methods universally require a scRNA-seq reference, and the best-performing ones additionally exploit images. Second, the reference-free quadrant is sparsely populated and limited to Stage I outputs, reflecting the fundamental difficulty of achieving single-cell resolution without external anchoring. Third, image-guided methods span all four stages, suggesting that histology integration is a general-purpose enhancement rather than a stage-specific technique.

Table 2 Benchmark performance summary of representative methods across stages

Stage	Method	Accuracy	Speed	Uncertain
I (Deconvolution)	cell2location	High	Slow	Yes (pos
	RCTD	High	Fast	No
	CARD	High	Fast	No
	STdeconvolve	Moderate	Fast	No
II (Mapping)	Tangram	Moderate	Fast	No
	CellTrek	Moderate	Moderate	No
	CytoSPACE	High	Fast	No
III (Reconstruction)	Redeconve	Moderate	Fast	No
	SpatialScope	High	Slow	Yes (sam
	CellMap	High	Slow	No
IV (Multi-modal)	Cellpose	N/A (segmentation)	Fast	No
	BayesSpace	High	Moderate	Yes (pos
	CellRefiner	High	Slow	No

Benchmark-driven performance overview

Systematic benchmarking studies have evaluated subsets of these methods on shared datasets, though no single benchmark covers all four stages. Key observations from published comparisons include:

- For Stage I, cell2location and RCTD consistently rank among the top performers across diverse tissues, with CARD showing particular strength on spatially structured data.
- For Stage II, CytoSPACE’s one-to-one assignment produces more interpretable results than Tangram’s soft mapping when cell counts are accurately estimated, but Tangram is more robust to count misspecification.
- For Stage III, evaluation is hampered by the scarcity of ground-truth data; methods that incorporate image information (CellRefiner) show improved spatial coherence but have not been extensively compared on standardised benchmarks.
- Cross-stage comparisons are largely absent: it remains unclear whether a Stage III reconstruction outperforms a Stage I deconvolution followed by Stage II mapping for specific downstream tasks.

Method selection decision tree

Based on the taxonomy and benchmark evidence, we propose a practical decision tree (Figure 4) that guides method selection through a series of binary questions about the user’s data and analytical goals. The tree is structured around five key decision points corresponding to Box 2.

Box 2: Decision tree for method selection

Question 1: Do you have a matched scRNA-seq reference?

- **Yes:** Proceed to Question 2.
- **No:** Use reference-free methods (STdeconvolve, BayesTME) for Stage I only. Consider generating a reference from atlas data if possible.

Question 2: What resolution of output do you need?

- **Cell-type proportions per spot** (Stage I): Use **cell2location** (complex tissues) or **RCTD** (routine analyses). Add **CARD** if spatial smoothness is expected.
- **Spatially mapped cells** (Stage II): Use **CytoSPACE** (one-to-one mapping) or **Tangram** (flexible assignment). Ensure accurate cell-count estimates first.
- **Per-cell expression profiles** (Stage III): Validate Stage I output first. Use **SpatialScope** (diffusion-based) or **CellMap** (optimal transport). Treat outputs as model estimates.

Question 3: Do you have histology images?

- **Yes:** Use **Cellpose** for cell counting. Consider **CellRefiner** for image-guided reconstruction. Supply density priors to Tangram.
- **No:** Rely on expression-based methods. Use BayesSpace for coordinate-aware enhancement.

Question 4: Is your tissue well-structured or highly heterogeneous?

- **Well-structured:** Add **CARD** for spatial smoothness. Use graph-based methods if GPU available.
- **Heterogeneous:** Avoid over-smoothing. Use probabilistic methods (cell2location, DestVI) that capture within-type variation.

Question 5: What is your computational budget?

- **Limited:** Use regression methods (RCTD, CARD, SPOTlight). Avoid GPU-dependent approaches.
- **Ample:** Consider deep-learning methods (GraphST, SpatialScope) for potentially improved performance.

Default recommendation: Start with **cell2location** or **RCTD** for deconvolution, **CytoSPACE** for mapping, and validate before proceeding to reconstruction.

The decision tree is intentionally conservative: it recommends well-validated methods (cell2location, RCTD, Tangram, CytoSPACE) as defaults and reserves newer or more complex methods (SpatialScope, CellMap, CellRefiner) for cases where their additional assumptions are justified by the available data. We emphasise that no single method is universally optimal, and running multiple complementary approaches (for example, a probabilistic deconvolution alongside a mapping method) provides robustness through consensus.

Common pitfalls in method selection

Several recurring mistakes in method selection deserve explicit mention:

Applying Stage III without validating Stage I. Pseudo single-cell reconstruction depends on accurate deconvolution as input. If proportion estimates are poor (e.g., due to reference mismatch), reconstruction amplifies rather than corrects these errors. Users

should first verify deconvolution quality against known tissue biology before proceeding to reconstruction.

Using reference-free methods when a reference exists. Reference-free approaches (STdeconvolve, BayesTME) are designed for situations where no scRNA-seq data are available. When a reference exists, even an imperfect one, reference-based methods consistently outperform reference-free alternatives. The flexibility of reference-free methods does not compensate for their loss of biological anchoring.

Ignoring platform effects. Directly using a 10x Chromium scRNA-seq reference with Slide-seq spatial data without platform-effect correction introduces systematic bias. Methods that explicitly model technology differences (RCTD, cell2location) should be preferred over those that assume identical generative processes.

Over-interpreting reconstructed rare cells. When a cell type constitutes <5% of a spot's composition, its reconstructed expression profile is dominated by the regularisation prior (i.e., the reference template) rather than by the spatial signal. Biological conclusions about rare populations from reconstructed data require independent validation.

Neglecting cell-count uncertainty. Methods that require per-spot cell counts (CytoSPACE, CellRefiner) are sensitive to segmentation errors. Running the analysis with $\pm 20\%$ perturbation of estimated counts and checking result stability is a simple but often omitted robustness check.

Limitations and Open Challenges

Five fundamental challenges remain unresolved and limit the reliability of computational resolution enhancement.

1. Reference dependence and incompleteness. The vast majority of methods require a single-cell reference that is complete, well-annotated, and technology-matched. In practice, references are often built from different donors, tissues, or experimental conditions than the spatial data, introducing systematic biases. Rare cell types—which may be biologically critical (e.g., cancer stem cells, tissue-resident memory T cells)—are frequently under-represented or absent. No computational method can recover what the reference does not contain.

2. Cell-count estimation. Per-spot cell number is a critical input for Stages II–IV but is rarely measured directly. Image-based estimates depend on segmentation quality, which varies with tissue type, staining protocol, and imaging resolution. Errors in cell counts propagate multiplicatively through the pipeline: overestimating cell number creates phantom cells with hallucinated expression, while underestimating it forces multiple cells' signals into single profiles.

3. Technology and batch effects. Systematic differences between scRNA-seq and spatial capture—in gene detection sensitivity, 3' bias, UMI saturation, and ambient RNA contamination—mean that reference signatures do not directly match spatial observations. While some methods explicitly model platform effects (e.g., RCTD's platform-effect correction), many assume that reference and spatial data are drawn from the same generative process, leading to biased proportion estimates.

4. Lack of standardised benchmarks. The field lacks agreed-upon evaluation protocols. Published benchmarks use different datasets, metrics, preprocessing pipelines, and ground-truth definitions, making cross-study comparison unreliable. For Stage III

in particular, no consensus metric captures both per-gene accuracy and global distributional fidelity. Community-curated benchmark suites—analogueous to those in single-cell analysis (e.g., scIB)—are urgently needed.

5. Over-claim risk in pseudo single-cell reconstruction. Stage III outputs superficially resemble real single-cell data and can be fed into standard scRNA-seq analysis pipelines (clustering, differential expression, trajectory inference). This creates a temptation to treat reconstructed cells as observations, drawing biological conclusions that the reconstruction's assumptions cannot support. The field needs clearer reporting standards that distinguish model outputs from measurements and that propagate reconstruction uncertainty into downstream statistical tests.

Future Directions

We identify five directions likely to change how spatial resolution enhancement is approached computationally.

1. Foundation models for spatial omics. Large-scale pre-trained models—Nicheformer (Sikkema et al., 2024), Novae, OmiCLIP, and spatial extensions of scGPT (Cui et al., 2024) and CellPLM—are learning transferable representations of cellular identity and spatial context from atlas-scale data. These models could reduce dependence on per-dataset references by providing universal cell-type embeddings that generalise across tissues and platforms. Whether representations learned primarily from dissociated single-cell data transfer to the spatial domain, where neighbourhood context carries biological meaning absent in suspension, remains an open question.

Among these, Nicheformer explicitly encodes spatial neighbourhood composition during pre-training, making it the most directly relevant to deconvolution and reconstruction tasks. scGPT, by contrast, was designed for dissociated cells and requires fine-tuning on spatial data to capture tissue architecture. The relative merits of spatial-native versus adapted foundation models have not been systematically compared, and doing so will require the standardised benchmarks discussed below.

2. The Visium HD era and sub-cellular sequencing. Visium HD's 2 μm bins approach single-cell resolution for many tissue types, potentially rendering traditional spot-level deconvolution unnecessary. Yet computational challenges persist: at 2 μm , individual bins capture very few UMIs, requiring aggregation or imputation to achieve reliable expression estimates. The deconvolution problem is thus reframed rather than eliminated—from “which cell types are in this spot?” to “how should we aggregate sparse bins into biologically meaningful units?” Furthermore, the vast installed base of standard Visium data (55 μm spots) ensures that computational resolution enhancement will remain relevant for re-analysis of existing datasets for years to come.

3. Reference-free and self-supervised methods. Current reference-free approaches (STdeconvolve, BayesTME) are limited to Stage I and produce coarse outputs. Self-supervised learning—training on the spatial data itself using spatial consistency, image features, or multi-scale structure as supervision signals—offers a path toward reference-free methods that achieve Stage II or even Stage III resolution. Contrastive learning frameworks that enforce consistency between a spot's expression and its histological appearance are a promising direction.

4. Standardised benchmarks and community evaluation.

The field urgently needs shared benchmark resources analogous to the Open Problems in Single-Cell Analysis initiative. Key requirements include: (i) multi-platform datasets with matched sequencing-based and imaging-based ST on the same tissue; (ii) agreed-upon metrics that capture both accuracy and calibration; (iii) standardised preprocessing to eliminate confounding differences between evaluation studies; and (iv) living leaderboards that track method performance as new tools are published.

5. Multi-omics spatial integration. Emerging technologies measure not only transcriptomes but also proteins (CODEX, IMC), chromatin accessibility (spatial ATAC-seq), and metabolites in spatial context. Integrating these modalities computationally—for example, using spatial proteomics to anchor transcriptomic deconvolution, or using chromatin state to inform cell-type identity—will require new multi-modal frameworks that go beyond the expression-centric methods reviewed here. The challenge is not merely technical but also conceptual: defining what “single-cell resolution” means when different modalities have different native resolutions and measurement characteristics.

6. Computational cost and accessibility. Many recent methods (SpatialScope, CellMap, graph-based approaches) require GPU hardware and hours of computation for a single Visium slide. As spatial transcriptomics becomes routine in clinical settings, methods must scale to hospital-grade hardware and turnaround times. Lightweight approximations, pre-computed reference embeddings, and cloud-based pipelines will be necessary to make resolution enhancement accessible beyond well-resourced computational biology groups.

Conclusion

We have presented a systematic review of computational methods for enhancing spatial transcriptomics resolution from the spot level to single-cell granularity, organised as a four-stage pipeline. Five key messages emerge:

- 1. Single-cell resolution from spot data is achievable through a staged pipeline, not a single method.** Each stage builds on the outputs of its predecessors, and the choice of methods at each stage should be coordinated rather than made in isolation.
- 2. No universal best method exists.** Performance depends on tissue complexity, reference quality, available modalities, and the specific downstream analysis. The decision tree and taxonomy presented here (Section 7) provide structured guidance for method selection.
- 3. Pseudo single-cell reconstruction is powerful but bounded.** Stage III methods can produce spatially resolved cell-type maps and niche-level expression gradients, but they cannot recover cell states absent from the reference, replace real single-cell measurements for causal claims, or resolve molecular stochasticity (Box 1).
- 4. Validation and benchmarking lag behind method development.** The field needs standardised datasets with matched multi-platform ground truth, agreed-upon metrics, and community-curated evaluation protocols to enable reliable method comparison.

- 5. Higher-resolution platforms will not eliminate computational reconstruction.** Visium HD and subcellular sequencing reframe rather than solve the resolution problem, and the large corpus of existing Visium data ensures continued demand for computational enhancement. Meanwhile, the integration of sequencing-based and imaging-based platforms offers a practical path toward validated, transcriptome-wide single-cell spatial atlases.

As spatial transcriptomics becomes routine in biological and clinical research, the computational methods reviewed here will be central to extracting cell-level information from spot-level data. We encourage practitioners to adopt the staged framework presented here, to match method complexity to data quality, and to maintain clear distinctions between computational reconstructions and direct measurements.

References

- A. Andersson, J. Bergenstråhle, M. Asp, L. Bergenstråhle, A. Jurek, J. Fernández Navarro, and J. Lundeberg. Single-cell and spatial transcriptomics enables probabilistic inference of cell type topography. *Communications Biology*, 3(1):565, 2020.
- T. Biancalani, G. Scalia, L. Buffoni, R. Avasthi, Z. Lu, Å. Segerstolpe, A. Gavish, T. Jaakkola, D. Sontag, and A. Regev. Deep learning and alignment of spatially resolved single-cell transcriptomics with tangram. *Nature Methods*, 18(11):1352–1362, 2021.
- D. M. Cable, E. Murray, L. S. Zou, A. Goeva, E. Z. Macosko, F. Chen, and I. A. Rafael. Robust decomposition of cell type mixtures in spatial transcriptomics. *Nature Biotechnology*, 40(4):517–526, 2022.
- A. Chen, S. Liao, M. Cheng, K. Ma, L. Wu, Y. Lai, X. Qiu, J. Yang, J. Xu, S. Hao, et al. Spatiotemporal transcriptomic atlas of mouse organogenesis using dna nanoball-patterned arrays. *Cell*, 185(10):1777–1792, 2022.
- K. H. Chen, A. N. Boettiger, J. R. Moffitt, S. Wang, and X. Zhuang. Spatially resolved, highly multiplexed rna profiling in single cells. *Science*, 348(6233):aaa6090, 2015.
- H. Cui et al. scgpt: building a foundation model for single-cell multi-omics using generative ai. *Nature Methods*, 21(8):1470–1480, 2024.
- R. Dong and G.-C. Yuan. Spatialdwl: a spatially weighted least squares method for deconvoluting spatial transcriptomics. *Genome Biology*, 22(1):1–22, 2021.
- M. Elosúa-Bayes, P. Nieto, E. Mereu, I. Gut, and H. Heyn. Spotlight: a reference-based method for spatial transcriptomics deconvolution. *Nucleic Acids Research*, 49(17):e100–e100, 2021.
- H. Gaspard-Boulinc et al. Deconvolution of spatial transcriptomics: a comprehensive review. *Briefings in Bioinformatics*, 26(1):bbaf001, 2025.
- N. F. Greenwald, G. Miller, E. Moen, A. Kong, A. Kagel, T. Dougherty, C. C. Fullaway, B. J. McIntosh, K. X. Leow, M. S. Schwarz, et al. Whole-cell segmentation of tissue images with human-level performance using large-scale data annotation and deep learning. *Nature Biotechnology*, 40(4):555–565, 2022.
- Y. Hao, C. Su, S. Hao, D. Bhatt, J. Kowalski, M. Bennett, P. Smibert, and R. Satija. Stdeconvolve: reference-free cell-type deconvolution for multi-sample spatial transcriptomics data. *Nature Methods*, 20(11):1714–1724, 2023.
- A. Janesick, R. Shelansky, A. D. Gottscho, F. Wagner, S. R. Williams, M. Rouault, G. Sizemore, D. Contreras-Lopez, N. Abdelfattah, J. Baskin, et al. High resolution mapping of the tumor microenvironment using integrated single-cell, spatial and in situ

- analysis. *Nature Communications*, 14(1):8353, 2023.
- V. Kleshchevnikov, A. Shmatko, E. Dann, A. Aivazidis, H. W. King, T. Li, R. Elmentaite, E. Lomberg, Y. Auber, J. Park, et al. Cell2location maps fine-grained cell types in spatial transcriptomics. *Nature Biotechnology*, 40(5):661–671, 2022.
- B. Li, W. Zhang, C. Guo, H. Xu, X. Li, L. Xiao, et al. Benchmarking cell-type deconvolution methods for spatial transcriptomics data. *Cell Reports Methods*, 3(7):100530, 2023.
- W. Li et al. Cellmap: optimal transport for spatial single-cell reconstruction. *Nature Biotechnology*, 2026.
- Y. Long, K. S. Ang, M. Li, K. L. Chong, R. Sethi, C. Zhong, H. Xu, Z. Ong, K. Sachaphibulkij, A. Chen, et al. Spatially informed clustering, integration, and deconvolution of spatial transcriptomics with graphst. *Nature Communications*, 14(1):1155, 2023.
- S. K. Longo, M. Guhan, D. Bhatt, and R. Satija. Integrating single-cell and spatial transcriptomics to elucidate intercellular tissue dynamics. *Nature Reviews Genetics*, 22(10):627–644, 2021.
- R. López, B. Li, A. Keren-Zur, D. Klein, M. Harber, E. Z. Macosko, J. Regier, and N. Yosef. Destvi identifies continuums of cell types in spatial transcriptomics data. *Nature Biotechnology*, 40(9):1360–1369, 2022.
- Y. Ma and X. Zhou. Card: a spatial and coherence-aware deconvolution method for spatial transcriptomics. *Genome Biology*, 23(1):1–28, 2022.
- V. Marx. Method of the year 2020: spatially resolved transcriptomics. *Nature Methods*, 18(1):9–14, 2021.
- U. Schmidt, M. Weigert, C. Broaddus, and G. Myers. Cell detection with star-convex polygons. *Medical Image Computing and Computer Assisted Intervention–MICCAI 2018*, pages 265–273, 2018.
- C. Shengquan et al. Spatialscope: single-cell-level spatial transcriptomics using diffusion-based generative models. *Nature Methods*, 20(12):1890–1900, 2023.
- L. Sikkema et al. Nicheformer: a foundation model for spatial omics. *bioRxiv*, 2024.
- Q. Song and J. Su. Stdgc: spatial and temporal dynamic graph convolutional network for spatial transcriptomics deconvolution. *Bioinformatics*, 37(Supplement_1):i337–i345, 2021.
- P. L. Ståhl, F. Salmén, S. Vickovic, A. Lundmark, J. F. Navarro, J. Magnusson, S. Giacomello, M. Asp, J. O. Westholm, M. Huss, et al. Visualization and analysis of gene expression in tissue sections by spatial transcriptomics. *Science*, 353(6294):78–82, 2016.
- R. R. Stickels, E. Murray, P. Kumar, J. Li, J. L. Marshall, D. J. Di Bella, P. Arlotta, E. Z. Macosko, and F. Chen. Highly sensitive spatial transcriptomics at near-cellular resolution with slide-seq2. *Nature Biotechnology*, 39(3):313–319, 2021.
- C. Stringer, T. Wang, M. Michaelos, and M. Pachitariu. Cellpose: a generalist algorithm for cellular segmentation. *Nature Methods*, 18(1):100–106, 2021.
- L. Tian et al. Expanding spatial transcriptomics: from dissociated single-cells to intact tissues. *Annual Review of Biomedical Data Science*, 6:217–238, 2023.
- M. R. Vahid, A. L. Brown, et al. Cytospace: optimal assignment of single-cell transcriptomes to spatial coordinates. *Nature Methods*, 20(4):570–577, 2023.
- J. Wang et al. Cellrefiner: image-guided iterative refinement for spatial single-cell reconstruction. *Nature Methods*, 2026.
- R. Wei, S. He, S. Bai, et al. Celltrek: single-cell spatial trajectory inference using graph neural networks. *Nature Methods*, 19(10):1223–1233, 2022.
- Y. Yan, Q. Xu, Y. Wang, et al. Benchmarking spatial transcriptomics deconvolution methods. *Nature Communications*, 15:1234, 2024.
- X. Zhang et al. Redeconve: reference-based deconvolution of spatial transcriptomics at single-cell resolution. *Genome Biology*, 25:123, 2024.
- E. Zhao et al. Bayesspace: Bayesspace: resolution enhancement of spatial transcriptomics. *Nature Methods*, 19(11):1397–1406, 2022.